

Practical 10:

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#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <string.h>

int main()
{
    const char *filename = "mmap_data.txt";

    int fd = open(filename, O_RDWR | O_CREAT, 0644);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    const char *initial_data =
        "This file is being accessed using mmap().\n";

    size_t size = strlen(initial_data);

    // Set the file size
    if (ftruncate(fd, size) == -1)
    {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    // Map the file into memory
    char *mapped = mmap(NULL,
                        size,
                        PROT_READ | PROT_WRITE,
```

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        perror("mmap");
        close(fd);
        return 1;
    }

    // Write initial data through mapped memory
    memcpy(mapped, initial_data, size);

    printf("Data read using mmap():\n");
    printf("%.s", (int)size, mapped);

    // Modify the file through memory
    const char *message = "Modified using mmap()!\n";

    size_t message_size = strlen(message);

    if (message_size <= size)
    {
        memcpy(mapped, message, message_size);
    }

    // Make changes visible to the file
    if (msync(mapped, size, MS_SYNC) == -1)
    {
        perror("msync");
    }

    printf("\nFile modified using mmap().\n");

    // Unmap memory
    munmap(mapped, size);

    close(fd);

    return 0;
}

```

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bhanu@DESKTOP-DE32HQ8:~$ cd ~
mkdir -p inode_mmap_lab
cd inode_mmap_lab
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ pwd
/home/bhanu/inode_mmap_lab
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ echo "This is an inode experiment." > original.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ cat original.txt
This is an inode experiment.
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ls -li original.txt
13073 original.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ stat original.txt
  File: original.txt
  Size: 29          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 13073        Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/   bhanu)   Gid: ( 1000/   bhanu)
Access: 2026-10-02 16:43:54.443629606 +0000
Modify: 2026-10-02 16:43:46.515798298 +0000
Change: 2026-10-02 16:43:46.515798298 +0000
Birth: 2026-10-02 16:43:46.515798298 +0000
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ln original.txt hardlink.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ls -li
total 8
13073 -rw-r--r--  2 bhanu bhanu 29 Oct  2 16:43 hardlink.txt
13073 -rw-r--r--  2 bhanu bhanu 29 Oct  2 16:43 original.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ echo "Data added through hard link." >> hardlink.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ cat original.txt
This is an inode experiment.
Data added through hard link.
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ rm original.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ls -li
total 4
13073 -rw-r--r--  1 bhanu bhanu 59 Oct  2 16:44 hardlink.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ echo "This is the symbolic link experiment." > original2.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ls -li original2.txt
1521 -rw-r--r--  1 bhanu bhanu 38 Oct  2 16:44 original2.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ln -s original2.txt symlink.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ls -li
total 8
13073 -rw-r--r--  1 bhanu bhanu 59 Oct  2 16:44 hardlink.txt
1521 -rw-r--r--  1 bhanu bhanu 38 Oct  2 16:44 original2.txt
1471 lrwxrwxrwx  1 bhanu bhanu 13 Oct  2 16:45 symlink.txt -> original2.txt

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13073 hardlink.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ find . -inum 123456
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ rm original2.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ls -l symlink.txt
lrwxrwxrwx 1 bhanu bhanu 13 Oct  2 16:45 symlink.txt -> original2.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ls -l symlink.txt
lrwxrwxrwx 1 bhanu bhanu 13 Oct  2 16:45 symlink.txt -> original2.txt
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ nano mmap_file.c
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ gcc -Wall -Wextra -o mmap_file mmap_file.c
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ./mmap_file
Data read using mmap():
This file is being accessed using mmap().

File modified using mmap().
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ ./mmap_file
Data read using mmap():
This file is being accessed using mmap().

File modified using mmap().
bhanu@DESKTOP-DE32HQ8:~/inode_mmap_lab$ |

```